COURSE OUTLINE (page 1)

Course: EE558, Optical Fiber Communication Systems

Objective: To obtain a familiarity with most major areas of optical communications as well as delve deeply into a few state-of-the-art research topics in the field.

Room: TV Center OHE 120
Hours: T, Th 9:30 - 11:00 am
Instructor: Prof. Alan Willner

Office: Rm. EEB 538, 213-740-4664, F:740-8729, willner@usc.edu, (H)310-839-3327

Office Hours: Tu: 11 am - 1 pm

Required Text:
1. Optical Fiber Communication Systems , Kazovsky, Benedetto, and Willner, Artech House

Other Suggested Texts: (To be put on reserve in Seaver Science and Engineering Library)
1. Optical Fiber Telecommunications I, Miller and Chynoweth, Academic Press
2. Optical Fiber Telecommunications II, Miller and Kaminow, Academic Press
3. Optical Fiber Telecommunications III, Kaminow and Koch, Academic Press
4. Optical Fiber Telecommunications IV, Kaminow and Li, Academic Press
5. Optical Fiber Telecommunications V, Kaminow, Li, and Willner, Academic Press
6. Optical Fiber Telecommunications VI, Kaminow, Li, and Willner, Academic Press
7. Introduction to Optical Fiber Communication Systems, W.B. Jones, HRW
8. Optical Communications Systems, Gowar, Prentice-Hall
9. Information, Transmission, Modulation, and Noise, Schwartz, McGraw Hill
10. Introduction to Optical Electronics, Yariv, HRW
11. Fiber Optic Communication, Killen, Prentice-Hall
12. Fiber Optic Communications, Palais, Prentice-Hall
13. Fiber Optic Communication Systems, Agrawal, Wiley

Grading:
- 40% - Midterm (covers Part I of next page)
- 5% - Homework
- 40% - Project (~50% written and ~50% oral)
- 15% - Mini-final (covers Part II of next page)

Project: To study a recent research topic in detail and clearly convey results to audience.

1) Topics to be chosen from a list after the first month of classes.

2) Project will require 2 consultations with instructor, first to discuss direction of topic and second for preview of presentation slides. These will be scheduled individually.

3) Oral presentations to begin after the midterm exam. 10-12 minutes plus discussion time per presentation. ****ALL students MUST present in class.****

4) Written presentation to be submitted near end of semester. 10-20 typed pages in length plus relevant references and figures.

EE558 COURSE OUTLINE (page 2)

I. Lectures on Major Areas of Optical Fiber Communications (~60% term)

A. Component Issues

1. Sources (multimode and singlemode lasers)
2. Fiber types, beam propagation, dispersion
3. Detectors and receivers
4. Connectors, couplers, isolators, polarization controllers
5. Amplifiers, filters, modulators

B. Systems Issues

1. Signal modulation formats and techniques
2. Detection schemes
3. Signal fidelity (signal-to-noise ratio and bit-error-rates)
4. Transmission and impairments
5. Optical multiplexing and switching (time, wavelength, and space)
6. Network switching, topologies and architectures
7. Network considerations (contention resolution)

II. Term projects on recent advances in Optical Fiber Communications

(complete list to be distributed in mid semester)

1. **Lasers** : micro-lasers, multiple-wavelength arrays, photonic-integrated-circuits
2. **Fiber** : dispersion-shifted, polarization-preserving, packaging, fiber-Bragg gratings
3. **Filters** : wavelength-tunable, bandwidth-tunable, wavelength-division multiplexers
4. **Receivers** : high-speed, arrays, optoelectronic integrated circuits
5. **Amplifiers** : semiconductor, Erbium-doped fiber, Raman, applications to long distance and distribution
6. **Switching**: electronic, space, time, and wavelength
7. **Long distance communications** : amplifier applications, solitons, dispersion, PMD, compensation, nonlinearities, and advanced modulation formats (PSK/QPSK/QAM, OFDM)
8. **Modulation and detection schemes**: direct, coherent
9. **Networks** : time-, space-, code-, and wavelength-division-multiplexing, multi-hop, circuit- and packet switching, contention resolution, network control and management
10. **Distribution systems** : fiber to the curb and/or home, hybrid fiber/coax, broadband networks, passive optical networks, SONET and ATM standards
11. **Subcarrier multiplexing** : FM, AM, digital, analog, cable-TV applications

- Each project will deal with a subset of each topic heading -